File

Name: P4-fit-A3-mapped-v2
Description: KISSsoft example
Changed by: AAA on: 30.10.2026 at: 19:37:00

# 1 Overview

Calculation method: According to DIN 7190-1:2017, elastic range
with additions such as centrifugal force, micro sliding, mounting, etc.
for a horizontal or longitudinal interference fit

| | | | | | | |
|---|---|---|---|---|---|---|
| Diameter shaft (mm) | [DiA]/[DiI] | 50.00 /10.00 | | | | |
| Diameter hub (mm) | [DaA]/[DaI] | 125.00 /50.00 | | | | |
| Length of Interference fit (mm) | [l] | 40.00 | | | | |
| Diameter of joint (mm) | [DF] | 50.00 | | | | |
| with special allowances: | | | | | | |
| Upper allowance Shaft (µm) | [AoI] | 73.9 | | | | |
| Lower allowance Shaft (µm) | [AuI] | 73.9 | | | | |
| Tolerance measure Shaft (µm) | [TI] | 0.0 | | | | |
| Upper allowance Hub (µm) | [AoA] | 0.0 | | | | |
| Lower allowance Hub (µm) | [AuA] | 0.0 | | | | |
| Tolerance measure Hub (µm) | [TA] | 0.0 | | | | |
| Largest interference (µm) | [Po] | 73.9 | | | | |
| Smallest interference (µm) | [Pu] | 73.9 | | | | |
| Nominal torque (Nm) | [T] | 0.00 | | | | |
| Application factor | [KA] | 1.00 | | | | |
| Service torque (Nm) | [Tb] | 0.00 | | | | |
| Axial force (N) | [FA] | 0.00 | | | | |
| Bending moment (Nm) | [Mb] | 0.00 | | | | |
| Radial force (N) | [Fr] | 0.00 | | | | |
| Circumferential force (N) | [Fu] | 0.00 | | | | |
| Speed (1/min) | [n] | 0.00 | | | | |
| Interference (µm) | [P] | 73.9 | ( 73.9 | .. | 73.9 | ) |
| Embedding (µm) | [s] | 4.80 | | | | |
| Effective interference at 20°C(68°F) (µm) | [Pw] | 69.14 | ( 69.1 | .. | 69.1 | ) |
| Effective interference at working temp.. (µm) | [PwTh] | 69.14 | ( 69.1 | .. | 69.1 | ) |
| Service temperature shaft (°C) | [ThB] | 20.0 | | | | |
| Service temperature hub (°C) | [ThB] | 20.0 | | | | |
| Pressure stress by | | | | | | |
| - Interference (after mounting) (N/mm²) | [pM] | 50.00 | ( 50.00 | .. | 50.00 | ) |
| - Interference (working) (N/mm²) | [p] | 50.00 | ( 50.00 | .. | 50.00 | ) |
| - Bending moment (N/mm²) | [pb] | 0.00 | | | | |
| - Radial force (N/mm²) | [pr] | 0.00 | | | | |
| Coefficient of friction, longitudinal | [µa] | 0.120 | | | | |
| Coefficient of friction, circumferential | [µu] | 0.120 | | | | |
| Safety against sliding | [Sr] | > 1000 | | | | |
| Required safety against sliding | [SSr] | 1.50 | | | | |

Equivalent stress according to von Mises
$\sigma V = (\sigma R^2+\tau T^2-\sigma R \cdot \tau T)^{0.5}$

## 1.1 Shaft

| | | | | | |
|---|---|---|---|---|---|
| Material, own Input | | C45 (1) | | | |
| Young's modulus (N/mm²) | [El] | 215000.00 | | | |
| Poisson's ratio (-) | [vl] | 0.30 | | | |
| Density (kg/m³) | [ρ] | 7830.00 | | | |
| Coefficient of thermal expansion (10^-6/K) | [α] | 11.50 | | | |
| Tensile strength (wall thickness) (N/mm²) | [Rm] | 650.00 | | | |
| Yield point (wall thickness) (N/mm²) | [Rp] | 300.00 | | | |
| The shaft's material strength values are defined using the wall thickness as the raw diameter. | | | | | |
| Component raw diameter (wall thickness) (mm) | [deff] | 21 | | | |
| | | | | | |
| Surface class of roughness | Own Input | | | | |
| Surface roughness (µm) | [Rz] | 6.00 | | | |
| External diameter (mm) | [DiA] | 50.00 | | | |
| Diameter increase during operation (µm) | [ΔD] | -9.11 | ( -9.11 | .. | -9.11 ) |
| Diameter increase during operation (o/oo) | [ΔD] | -0.18 | ( -0.18 | .. | -0.18 ) |
| Equivalent stress outer diameter (N/mm²) | [σVa] | 52.21 | ( 52.21 | .. | 52.21 ) |
| - Radial stress (N/mm²) | [σRa] | -50.00 | (-50.00 | .. | -50.00 ) |
| - Tangential stress (N/mm²) | [σTa] | -54.17 | (-54.17 | .. | -54.17 ) |
| - With outside load (N/mm²) | [σVaMF] | 52.21 | ( 52.21 | .. | 52.21 ) |
| Inner diameter (mm) | [Dil] | 10.00 | | | |
| Diameter increase during operation (µm) | [ΔD] | -4.84 | ( -4.84 | .. | -4.84 ) |
| Diameter increase during operation (o/oo) | [ΔD] | -0.48 | ( -0.48 | .. | -0.48 ) |
| Equivalent stress inner diameter (N/mm²) | [σVi] | 104.17 | (104.17 | .. | 104.17 ) |
| - Radial stress (N/mm²) | [σRi] | 0.00 | ( 0.00 | .. | 0.00 ) |
| - Tangential stress (N/mm²) | [σTi] | -104.17 | (-104.17 | .. | -104.17 ) |
| | | | | | |
| Safety against fracture | [SiRm] | 6.24 | ( 6.24 | .. | 6.24 ) |
| Required safety against fracture | [SSi.Rm] | 1.50 | | | |
| | | | | | |
| Safety against yield point | [Si.Rp] | 2.88 | ( 2.88 | .. | 2.88 ) |
| Required safety against yield point | [SSi.Rp] | 1.20 | | | |

**1.2 Hub**

| | | | | | |
|---|---|---|---|---|---|
| Material, own Input | | C45 (1) | | | |
| Young's modulus (N/mm²) | [EA] | 70000.00 | | | |
| Poisson's ratio (-) | [vA] | 0.30 | | | |
| Density (kg/m³) | [ρ] | 7830.00 | | | |
| Coefficient of thermal expansion (10^-6/K) | [α] | 11.50 | | | |
| Tensile strength (wall thickness) (N/mm²) | [Rm] | 700.00 | | | |
| Yield point (wall thickness) (N/mm²) | [Rp] | 150.00 | | | |
| The hub material strength values are defined using the wall thickness as the raw diameter. | | | | | |
| Component raw diameter (wall thickness) (mm) | [deff] | 40 | | | |
| | | | | | |
| Surface class of roughness | Own Input | | | | |
| Surface roughness (µm) | [Rz] | 6.00 | | | |
| External diameter (mm) | [DaA] | 125.00 | | | |
| Diameter increase during operation (µm) | [ΔD] | 34.01 | ( 34.01 | .. | 34.01 ) |
| Diameter increase during operation (o/oo) | [ΔD] | 0.27 | ( 0.27 | .. | 0.27 ) |
| Equivalent stress outer diameter (N/mm²) | [σVa] | 19.05 | ( 19.05 | .. | 19.05 ) |
| - Radial stress (N/mm²) | [σRa] | 0.00 | ( 0.00 | .. | 0.00 ) |
| - Tangential stress (N/mm²) | [σTa] | 19.05 | ( 19.05 | .. | 19.05 ) |
| Inner diameter (mm) | [Dal] | 50.00 | | | |
| Diameter increase during operation (µm) | [ΔD] | 60.03 | ( 60.03 | .. | 60.03 ) |
| Diameter increase during operation (o/oo) | [ΔD] | 1.20 | ( 1.20 | .. | 1.20 ) |
| Equivalent stress inner diameter (N/mm²) | [σVi] | 103.54 | (103.54 | .. | 103.54 ) |
| - Radial stress (N/mm²) | [σRi] | -50.00 | (-50.00 | .. | -50.00 ) |
| - Tangential stress (N/mm²) | [σTi] | 69.05 | ( 69.05 | .. | 69.05 ) |

| | | | | | |
|---|---|---|---|---|---|
| - Radial stress with outside load (N/mm²) | [σRai] | -50.00 | (-50.00 .. | -50.00 ) |
| - Tangential stress with outside load (N/mm²) | [σTai] | 69.05 | ( 69.05 .. | 69.05 ) |
| - With outside load (N/mm²) | [σViMF] | 103.54 | (103.54 .. | 103.54 ) |
| | | | | | |
| Safety against fracture | [Si.Rm] | 6.76 | ( 6.76 .. | 6.76 ) |
| Required safety against fracture | [SSi.Rm] | 1.50 | | |
| | | | | | |
| Safety against yield point | [Si.Re] | 1.45 | ( 1.45 .. | 1.45 ) |
| Required safety against yield point | [SSi.Re] | 1.20 | | |

Mere elastic stress, no verification of elastic plastic interference fit according to DIN 7190-1.

## 2 Service / Mounting / Dismounting

| | | | | | |
|---|---|---|---|---|---|
| Transverse-interference-fit: | | | | | |
| Mounting clearance (mm) | [PsTh] | 0.050 | | | |
| Temperature difference for mounting: | | | | | |
| Shaft temperature: (°C) Hub temperature: [ThA] (°C) | | | | | |
| Mount with shaft temperature 20 °C: | | | | | |
| Assembly temperature shaft (°C) | [θI] | 20 | | | |
| Assembly temperature hub (°C) | [θA] | 235.552 | | | |
| Mount with shaft temperature -150 °C: | | | | | |
| Assembly temperature shaft (°C) | [θI] | -150 | | | |
| Assembly temperature hub (°C) | [θA] | 109.900 | | | |
| calculated using the coefficient of thermal expansion | | | | | |
| Sub-cooling of the shaft according to DIN 7190-1 (10^-6/K) | [αw] | 8.50 | | | |
| | | | | | |
| Longitudinal pressure fit: | | | | | |
| Assembly temperature shaft (°C) | [θM] | 20.00 | | | |
| Assembly temperature hub (°C) | [θM] | 20.00 | | | |
| Coefficient of friction, longitudinal | [μe=μa* | 1.300 ] | 0.16 | | |
| Press on (force) (kN) | [Fpress] | 49.01 | ( 49.01 .. | 49.01 ) |
| Coefficient of friction, longitudinal | [μll=μa*1.600 ] | 0.19 | | | |
| Press out (force) (kN) | [Fpress] | 60.32 | ( 60.32 .. | 60.32 ) |
| | | | | | |
| Coefficient of friction | [μ] | 0.19 | | | |
| Max. torque to avoid Micro sliding (Nm) | [Tlimit] | 305.37 | (305.37 .. | 305.37 ) |

Notice concerning the display: Number-1 (Number-2.. Number-3):

Number-1: Value calculated with the mean allowance

Number-2: Value of the smallest possible allowance

Number-3: Value of the largest possible allowance

Notice: All strains and stresses are calculated for the purely elastic case.

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End of report (lines: 157)

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